

Zhenfei Wang

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RESEARCH INTERESTS

Computational linguistics | Cross-lingual NLP | Low-resource machine translation | Prompt engineering for linguistically texts | Corpus linguistics

EDUCATION

Guangdong University of Technology (GDUT) | Guangzhou, China

Bachelor of Arts in English (First Cohort, "English + AI" Innovation Program)

Minor: Computer Science | 09/2023 – 07/2027 (Expected)

• GPA (English Major): 87.4/100 | Overall GPA: 86.6/100

• Core Modules: Natural Language Processing (90), Machine Learning (90), Algorithm Design & Analysis (96), Artificial Intelligence (100), Corpus Linguistics (91), Database Systems (92), Big Data and Visualization (92), Computer Networks (93), Digital Media Technology (95), Database Systems Practicum (95), Data Structures, Computer Organization, Operating Systems

RESEARCH EXPERIENCE

Research Intern, Speech and Language Laboratory, Nanyang Technological University, Singapore (Remote)

07/2026 – Present

- Conducting research on the synthesis and automatic evaluation of non-verbal vocalizations (e.g., laughter, sighs) for TTS systems, supervised by Prof. Chng Eng Siong.
- Building an LLM-driven data pipeline: generating text data with LLMs, synthesizing speech containing diverse non-verbal cues with speech LLMs, and verifying outputs through an LLM-as-a-judge protocol.
- Curating a high-quality synthetic speech dataset via dual-agreement filtering—retaining only audio samples independently validated by two LLM judges.

Research Project, "RAG-based Continuation–Output Binding Mechanism for Writing Instruction"

12/2025 – Present

- Investigated how Retrieval-Augmented Generation (RAG) can be integrated with the continuation–output hypothesis to enhance AI-assisted writing instruction and address the lack of mechanism-level validation in AI-supported language learning.
- Independently designed the research framework (Coze), integrating SLA theory with RAG architecture; developed hypotheses, experimental design, corpus preparation plan, prompt tasks, and evaluation metrics for writing outcomes.
- Established the research framework and technical pathway, with ongoing data preparation and experimental implementation for theory-driven evaluation of RAG-based writing support.

Research Project, "Prompt Optimization for Chinese Classical Poetry Translation Based on Large Language Models"

11/2025 – Present

- Examined how prompt engineering can improve LLM-based translation of Chinese classical poetry, addressing the lack of theory-driven prompt design for linguistically and culturally complex texts.
- Based on Xu Yuanchong's translation theories, collected 7 poems, designed 9 structured prompt architectures and translate them across multiple Chinese and international LLMs (DeepSeek-V3.1, Kimi-K2, ChatGPT) (486 prompt-translation pairs). Led research design, corpus preparation, experimental workflow (Coze) construction, and data analysis, with translation outputs assessed through expert scoring and SPSS (Cronbach's $\alpha = .789$) with a self-made three-dimension evaluation scale.
- Found the main effect of prompt framework on translation quality ($p < .001$), completed the core experimental framework and drafted a manuscript of approximately 13,000 words, which is currently undergoing refinement and optimization before submission.

Research Project, "Research Hotspots and Visualized Analysis of Prosodic Grammar Based on VOSviewer and CiteSpace" (Provincial Undergraduate Innovation Project, Team Leader)

11/2025 – Present

- Investigated the intellectual structure, research hotspots, and emerging trends in prosodic grammar, focusing on the scholarly contributions of Feng Shengli, addressing the lack of systematic bibliometric synthesis in this field.
- Independently conducted research design, literature collection, data cleaning, and bibliometric analysis using CiteSpace and VOSviewer, analyzing over 5,000 citation records and managing large-scale literature datasets with tools such as Google Scholar, CNKI, Zotero, and NoteExpress.
- Completed a manuscript of over 10,000 words; currently under revision with supervisor guidance and planned for submission to the *SSCI*.

Research Project, "Decoding Cultural Codes: An Analysis of English Subtitle Translation in *Ne Zha: Birth of the Demon Child*" (Undergraduate Innovation Project)

11/2024 – 12/2025

- Examined subtitle translation strategies in the Chinese animated film *Ne Zha: Birth of the Demon Child* from the perspective of Skopos Theory, focusing on how culturally specific elements are conveyed to international audiences.
- Collaborated with four team members under supervision to analyze subtitle translation cases. Contributed to literature review, data collection, theoretical analysis, and the writing of the final research paper.
- Successfully completed the project and produced a final research paper, strengthening my skills in theoretical analysis, academic writing, and collaborative research.

Research Project, "Cultural Discount in the Cross-Cultural Communication of Chinese Domestic Films" (Undergraduate Innovation Project, Team Leader)

11/2023 – 06/2024

- Investigated the phenomenon of cultural discount in the international dissemination of Chinese films, examining how cultural specificity affects cross-cultural audience reception.
- Served as team leader, coordinating two members under supervision to conduct a literature review on cultural discount and cross-cultural film communication, including database searches and analysis using CNKI and Web of Science.

- Completed a final literature review report, strengthening my skills in academic writing, literature synthesis, and collaborative

ACADEMIC PROJECTS

Exchange Program (The Hong Kong Polytechnic University × GDUT), "Scoop One: Branding Upgrade Plan for KUKU Ice Cream Robot", Credit-based GBA Immersive Experience Program (3rd Cohort) 07/2026

- Selected for the 8-day, credit-bearing "AI + Robotics" program jointly offered by PolyU and GDUT (42 students from 20+ majors), with English-medium Design Thinking coursework and credits mutually recognized by both universities.
- Conducted field research at GBA robotics enterprises (KUKU Robotics, Aican Technology, Hikvision); worked in a 6-member cross-university team on a real challenge posed by KUKU Robotics—transforming its ice cream robot from a vending machine into a child-oriented interactive brand experience.
- Led user research and needs analysis (children aged 3–12 and parents) and presented the "Customer Opportunity" module in the final business pitch; co-proposed a low-cost modular retrofit (IP stickers, logo upgrade, joystick and light-button modules) with a phased MVP pilot plan (3 machines, 4–6 weeks, ¥300,000 budget cap), and built one of the team's two AI-assisted 3D prototype models.

Course Project (Big Data and Visualization), "Food Loanword Borrowing under Globalization: An Interactive Data Visualization Study" (Team Leader) 06/2026

- Led a team to build an interactive web-based research report on 662 deduplicated English food loanwords, integrating Google Books Ngram frequencies (1800–2020) with Merriam-Webster first-attestation and inclusion data.
- Collected and cleaned multi-source lexical data (828 raw entries → 662 words) in Python; conducted Pearson chi-square tests in SPSS (dictionary inclusion × semantic category) and machine learning analysis in scikit-learn (K-Means on frequency trajectories; Random Forest on peak frequency).
- Proposed a "dual-track asymmetry" framework contrasting corpus frequency with dictionary inclusion, revealing stratified waves of loanword influx across 13 source languages.
- Designed and deployed an interactive visualization website (React, Vite, ECharts) on GitHub Pages: willowwanglala-glitch.github.io/globalization-food-words

Course Project (Machine Learning), "Bilingual Corpus Construction and Machine Translation Model Training for Chinese Political Discourse Based on LSTM". Undergraduate Course Paper

- Training an English translation model for Chinese political documents based on an LSTM architecture.
- Constructed a self-built Chinese–English parallel corpus of approximately 320,000 words covering Chinese political documents such as government work reports and white papers. After text cleaning, tokenization, and vocabulary construction, a three-layer LSTM neural network was implemented using PyTorch. The model was trained and evaluated with an 80/20 train–test split.
- The training process was stable, with the loss value decreasing steadily and converging at 8.3047. The results demonstrate the feasibility of LSTM-based architectures for machine translation of political discourse and provide empirical reference for translation modeling in low-resource or domain-specific settings.
- Strengthened my skills in bilingual corpus construction, neural machine translation modeling, and PyTorch-based implementation, and deepened my understanding of sequence modeling and domain-adapted machine translation.

Course Project (Corpus Linguistics), "Gendered Discourse Analysis of Subtitles from Top 30 Global Box Office Films in 2025". Undergraduate Course Paper

- Analysis of gender-related discourse differences in the Top 30 global box office films (2025).
- Constructed a target corpus of subtitles from 14 popular films, comprising 138,137 words. Used the LOB and Brown corpora as reference corpora. Using tools such as AntConc, WordSmith, and SPSS, conducted keyword analysis (e.g., *he/she*, *man/woman*), log-likelihood (LL) value calculation, and part-of-speech tagging to examine gendered discourse patterns.
- Exhibited clear gendered discourse differences in the film subtitle corpus, with male-associated language more frequently linked to power and action, and female-associated language more often related to appearance and family. Statistically significant differences were observed between the film corpus and the reference corpora. The constructed corpus also demonstrates potential value for English learning and machine translation training.
- Strengthened skills in corpus construction, quantitative discourse analysis, and statistical testing, and enhanced ability to apply corpus linguistics methods to sociolinguistic and NLP-related research tasks.

Course Project (Software Engineering), "Language for Life: AI-Powered English Exam Prep Mini Program"

- Team Leader & Backend/AI Lead. Led 2-member team through 4-Sprint agile development, coordinating cross-functional integration and deliverable scheduling
- Architected unified backend API gateway (20 cloud functions) with MongoDB schema design and dual-mode cloud/local deployment
- Orchestrated Coze AI workflows for intelligent tutoring, study plan generation, and essay evaluation with fault-tolerant degradation pipelines
- Authored system architecture and database design specifications; managed GitHub repository, release management, and defense deliverables
- Applied prompt engineering and fault-tolerant AI pipeline design in an educational technology context, extending research expertise in AI-assisted language learning from theoretical frameworks to full-stack product implementation.

AWARDS & HONORS

- Second Prize (Provincial), "Telling China's Stories Well in Foreign Languages" Short Video Competition, Guangdong Province 2025
- Bronze Award (Provincial), FLTRP · iTest Cup Short Video Competition, Guangdong Province 2024
- Bronze Award (University), FLTRP · iTest Cup Translation Competition (Written Translation), Guangdong University of Technology 2025
- Merit Award (University), National College Students New Liberal Arts Practice and Innovation Competition 2024
- First-Class Academic Excellence Scholarship, Guangdong University of Technology 2023–2024
- Second Prize (University), FLTRP · iTest Cup Reading Competition (English Group), Guangdong University of Technology 2023
- Outstanding Team Leader, "Winter Break Alumni Outreach Program" for Admissions Promotion, Guangdong University of Technology 2025

•Gold Award (University), FLTRP · iTest Cup Short Video Competition, Guangdong University of Technology	2024
•Outstanding Team Leader, Winter Break Alumni Outreach Program for Admissions Promotion	2025
•Merit Award (School), National College Students New Liberal Arts Practice and Innovation Competition	2024
•Outstanding Orientation Ambassador, Guangdong University of Technology	2023-2024
•Advanced Individual in Communist Youth League Work, Guangdong University of Technology	2023–2024
•Outstanding Communist Youth League Member, Guangdong University of Technology	2023

CERTIFICATIONS

- **English:** CET-4(569), CET-6(523); TEM-4 (Good); IELTS 6.0 (Overall) , retake scheduled in July 2026 targeting 6.5+
- **Mandarin Chinese:** Level 2-B

SKILLS

- **Programming:** C, Python
- **Tool:** AntConc, WordSmith, SPSS, CiteSpace, Coze, VOSviewer
- **Design:**123 Editor, Mageda, Adobe Photoshop, PPT, Canva, Powerpoint, Media (11 articles and 3 video, official Wechat account “GDUT Youth Hub”, 30k+ reach)
- **Language:** Mandarin Chinese (Native); Hakka (Native); English (Proficient); Spanish (Intermediate)

ADDITIONAL INFORMATION

- **Leadership:** Class academic committee member; Winter admissions outreach team leader;
- **Entrepreneurship:** Co-founded “Waimai Go” studio at GDUT Innovation Hub (2024)
- **Public Service:** Administrative assistant, Country Education Bureau (2025)
- **Design:** “GDUT Youth Hub” member